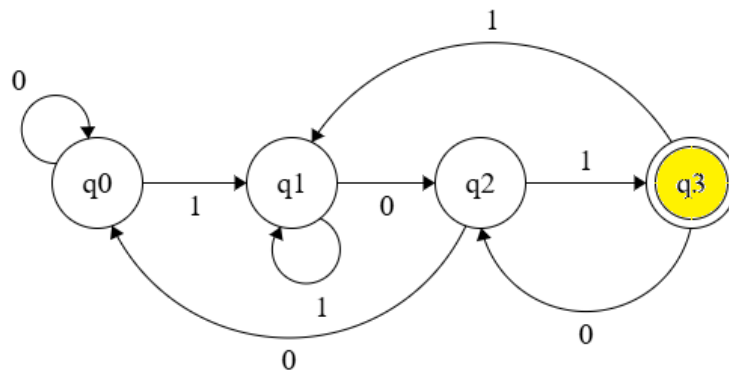


Practical No.5

Aim: Design a Program for creating machine that accepts the string always ending with 101.



Program Code:

```
practical_no.5.py - H:\SYCS TOC\SYCS TOC practical\practical_no.5.py (3.12.0)
File Edit Format Run Options Window Help
1 '''Pract5:Design a Program for creating machine that
2 accepts the string always ending with 101.
3 '''
4 def stateq0(n):
5     print("q0->",end="")
6     if (len(n)==0):
7         print("exit")
8         print("String is rejected")
9
10    else:
11        if (n[0]=='0'):
12            stateq0(n[1:])
13
14        elif (n[0]=='1'):
15            stateq1(n[1:])
16
17 def stateq1(n):
18     print("q1->",end="")
19     if (len(n)==0):
20         print("exit")
21         print("String is rejected")
22    else:
23        if (n[0]=='0'):
24            stateq2(n[1:])
```

```

25
26     elif (n[0]=='1'):
27         stateq1(n[1:])
28
29 def stateq2(n):
30     print("q2->",end="")
31     if (len(n)==0):
32         print("exit")
33         print("String is rejected")
34
35     else:
36         if (n[0]=='0'):
37             stateq0(n[1:])
38
39         elif (n[0]=='1'):
40             stateq3(n[1:])
41
42
43 def stateq3(n):
44     print("q3->",end="")
45     if (len(n)==0):
46         print("exit")
47         print("String is accepted")
48     else:
49         if (n[0]=='0'):
50             stateq2(n[1:])
51
52         elif (n[0]=='1'):
53             stateq1(n[1:])
54
55 #take number from user
56 num=(input("Please enter any one binary number:\n"))
57 #call stateA to check the input
58 stateq0(num)
59

```

Output:

```

Python 3.12.0 (tags/v3.12.0:0fb18b0, Oct 2 2023, 13:03:39) [MSC v.1935 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: H:/SYCS TOC/SYCS TOC practical/practical_no.5.py
Please enter any one decimal number:

===== RESTART: H:/SYCS TOC/SYCS TOC practical/practical_no.5.py =====
Please enter any one binary number:
10101010
q0->q1->q2->q3->q0->q1->q2->q3->q0->exit
String is rejected
>>>

===== RESTART: H:/SYCS TOC/SYCS TOC practical/practical_no.5.py =====
Please enter any one binary number:
111111101101
q0->q1->q1->q1->q1->q1->q1->q1->q2->q3->q1->q2->q3->exit
String is accepted
>>>

```